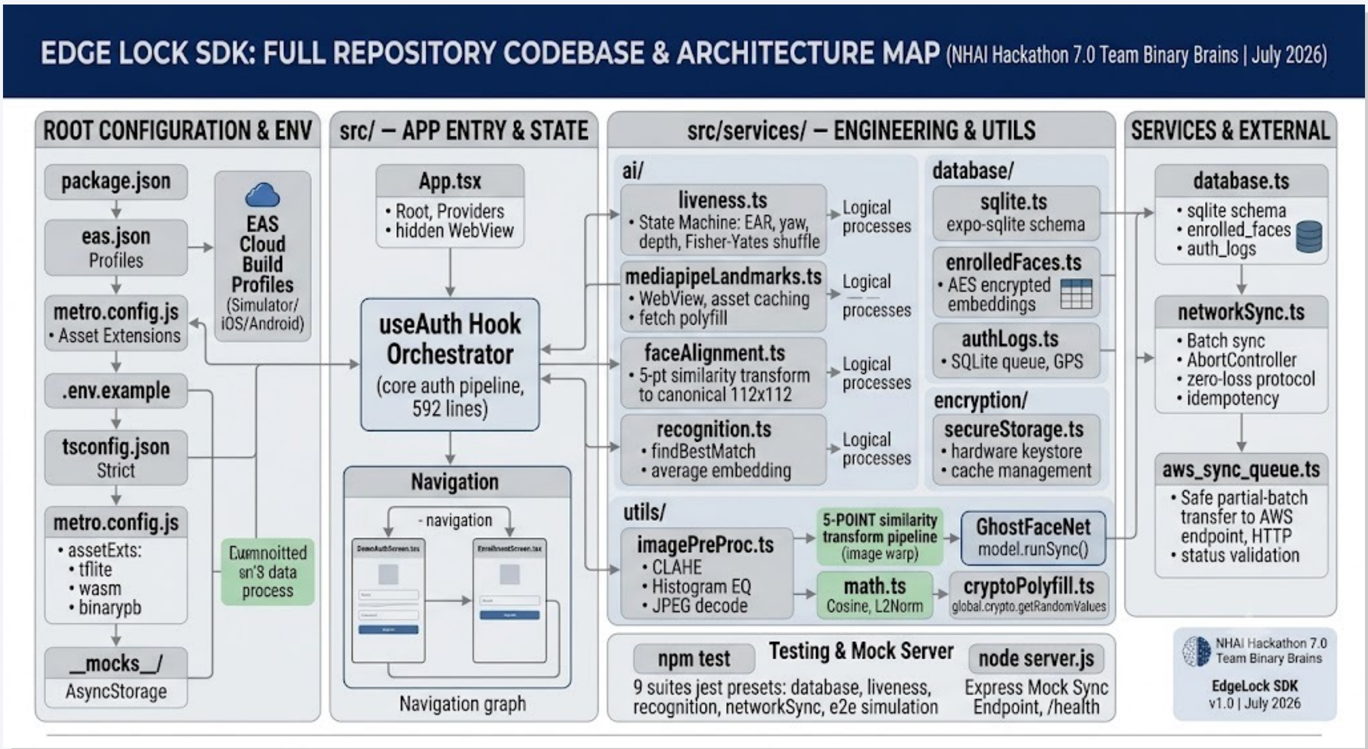
EDGE LOCK SDK

Visual Technical Deck

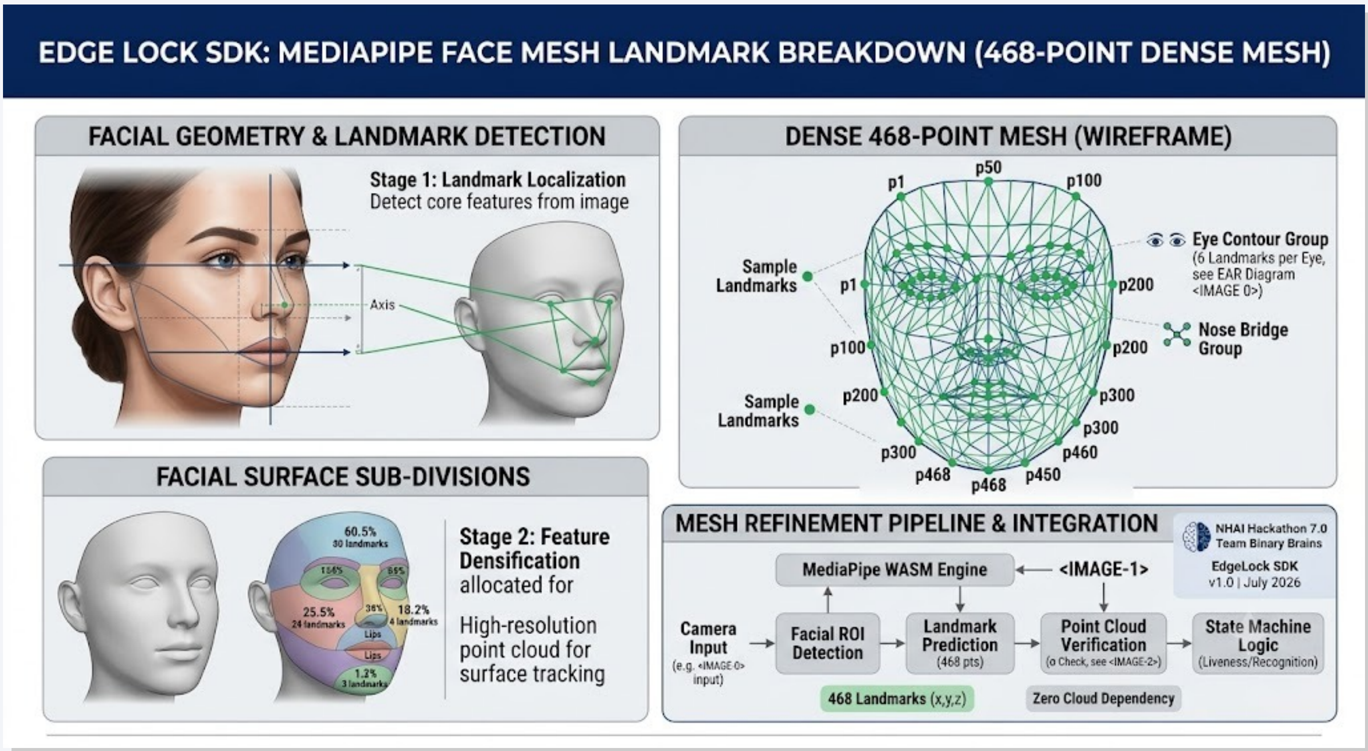
NHAI Hackathon 7.0 | Team Binary Brains | July 2026

Offline Facial Recognition with Active Liveness & Zero-Loss Sync

SYSTEM OVERVIEW

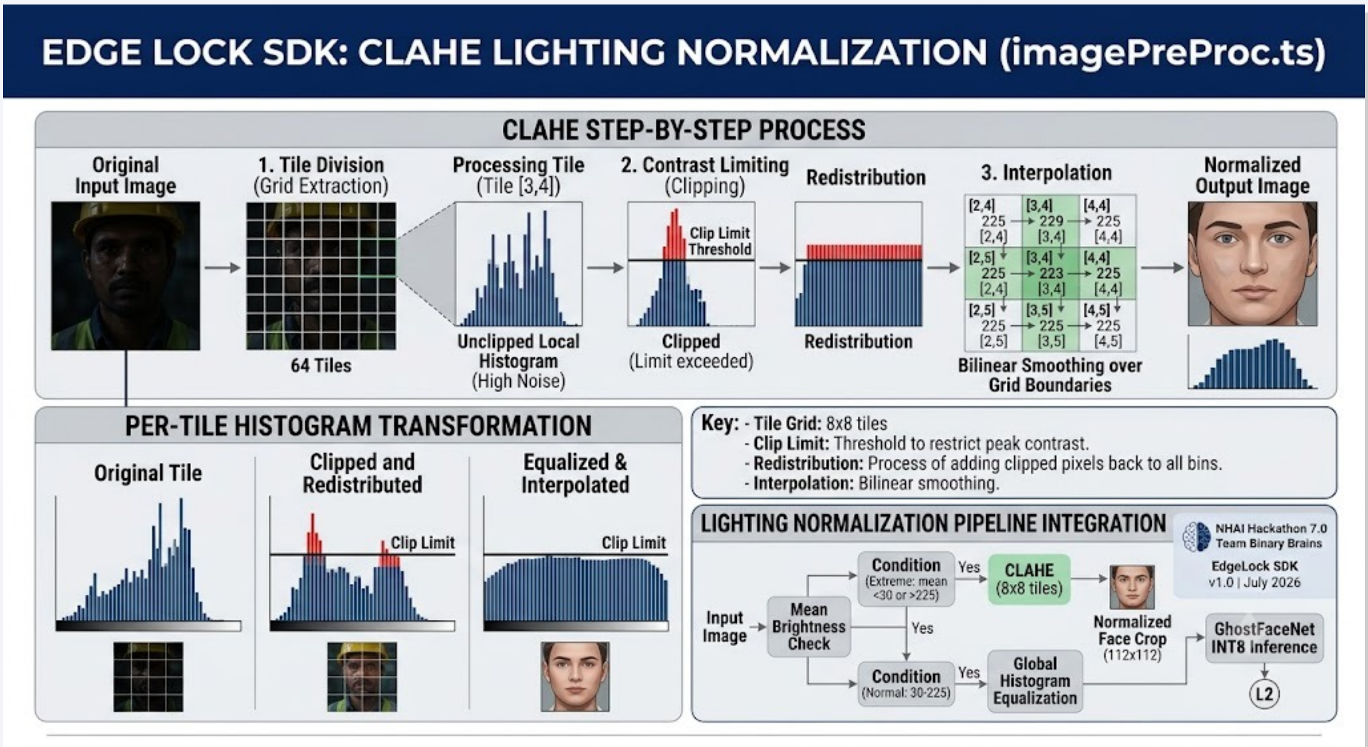


Full Repository Architecture & Codebase Map

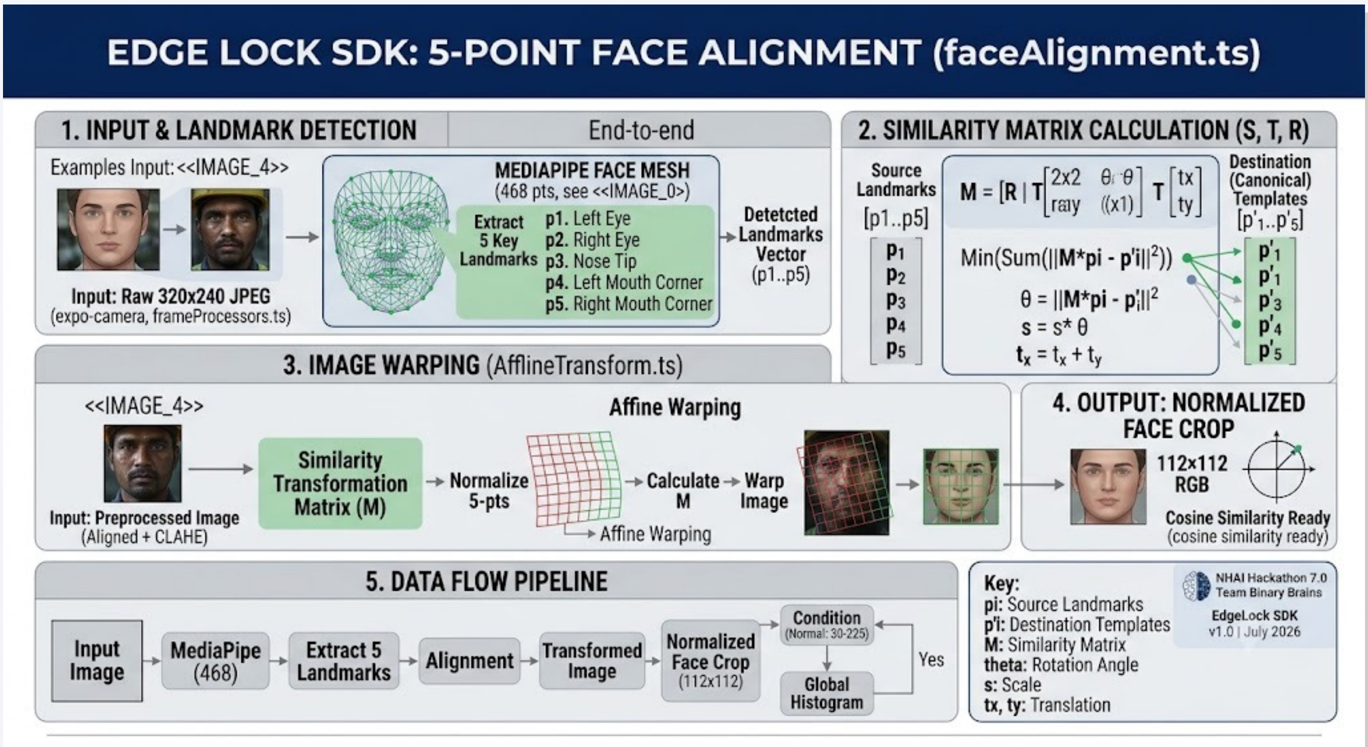


MediaPipe Face Mesh — 468-Point Dense Landmark Detection

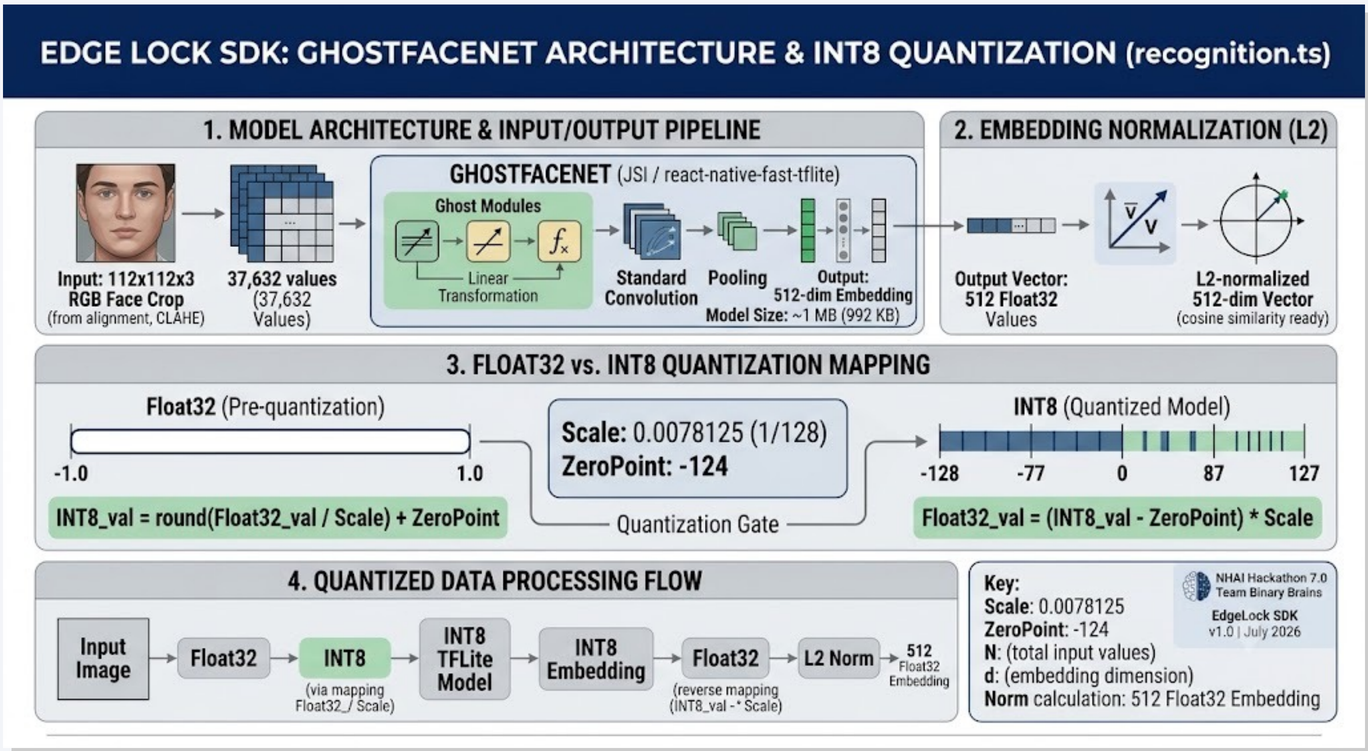
PREPROCESSING



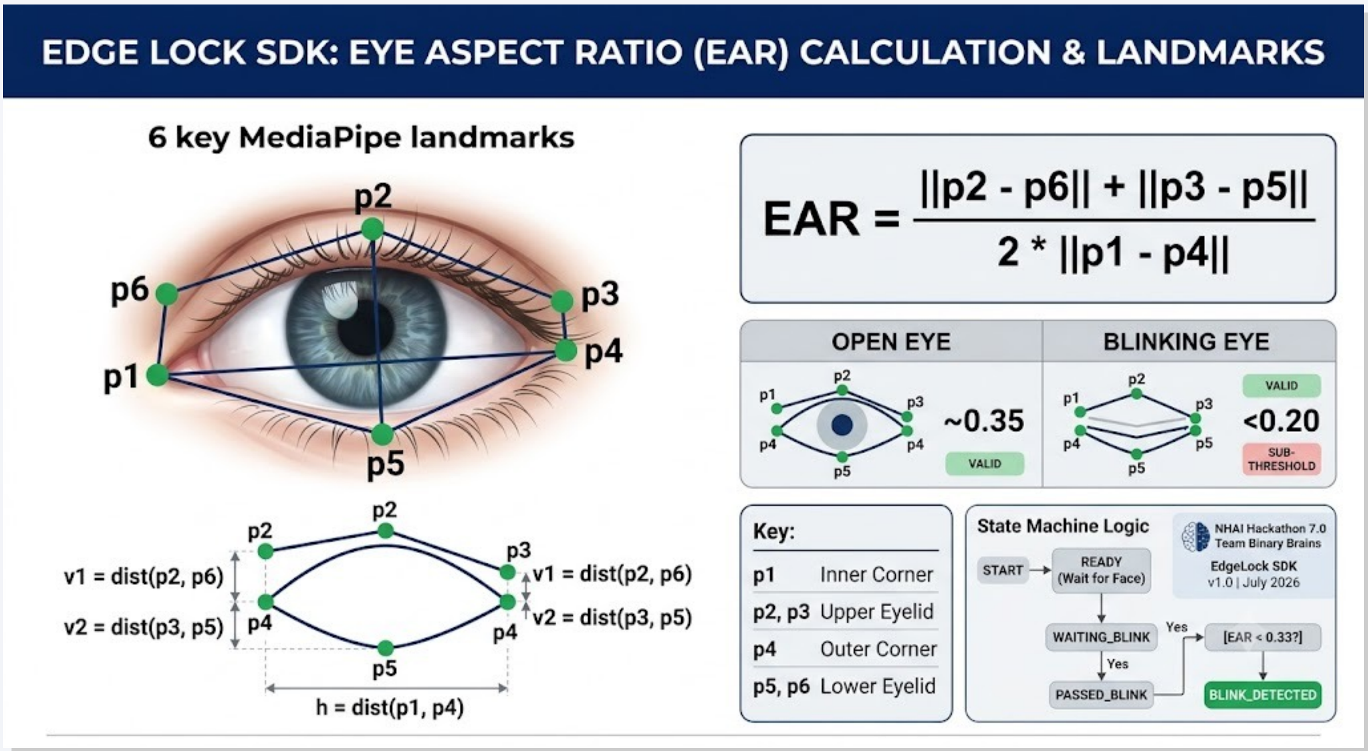
CLAHE Lighting Normalization — Harsh Field Conditions



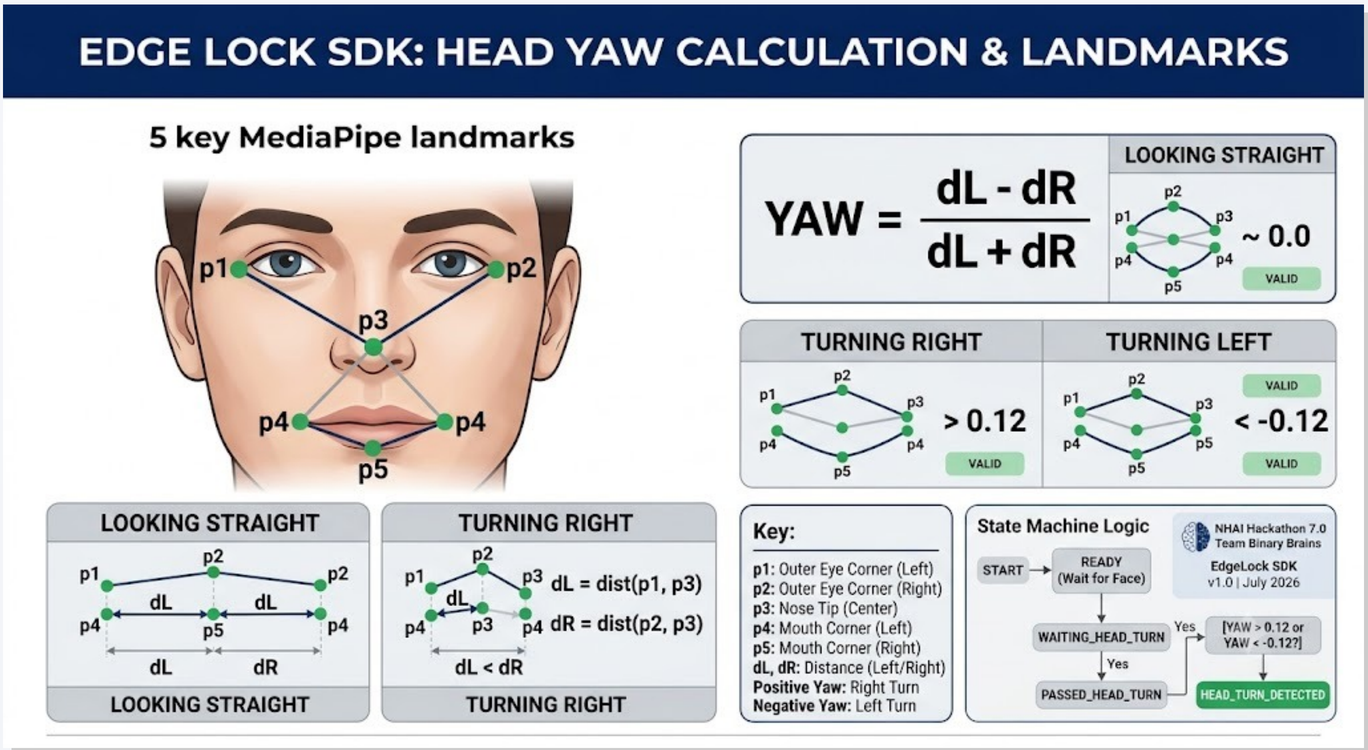
5-Point Face Alignment — Affine Warp to Canonical Crop



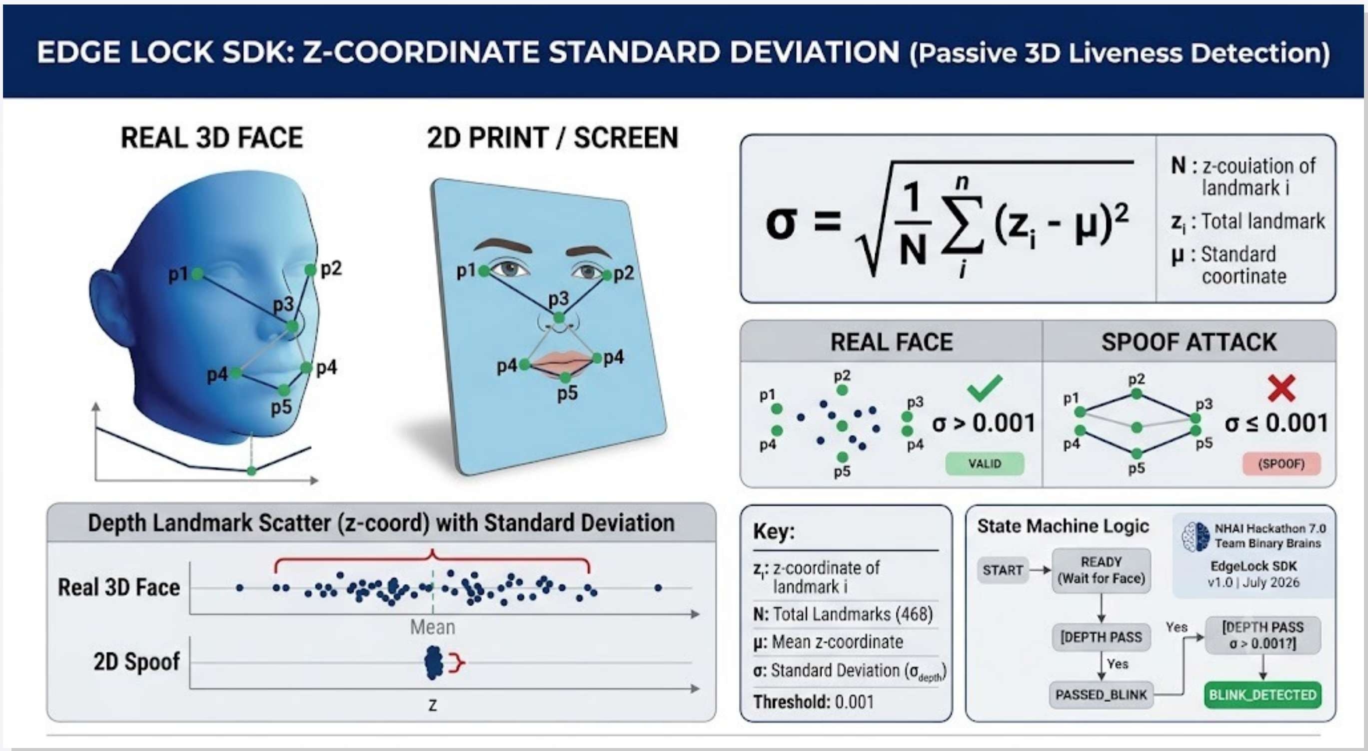
GhostFaceNet INT8 Quantized — 512-dim Embedding Extraction



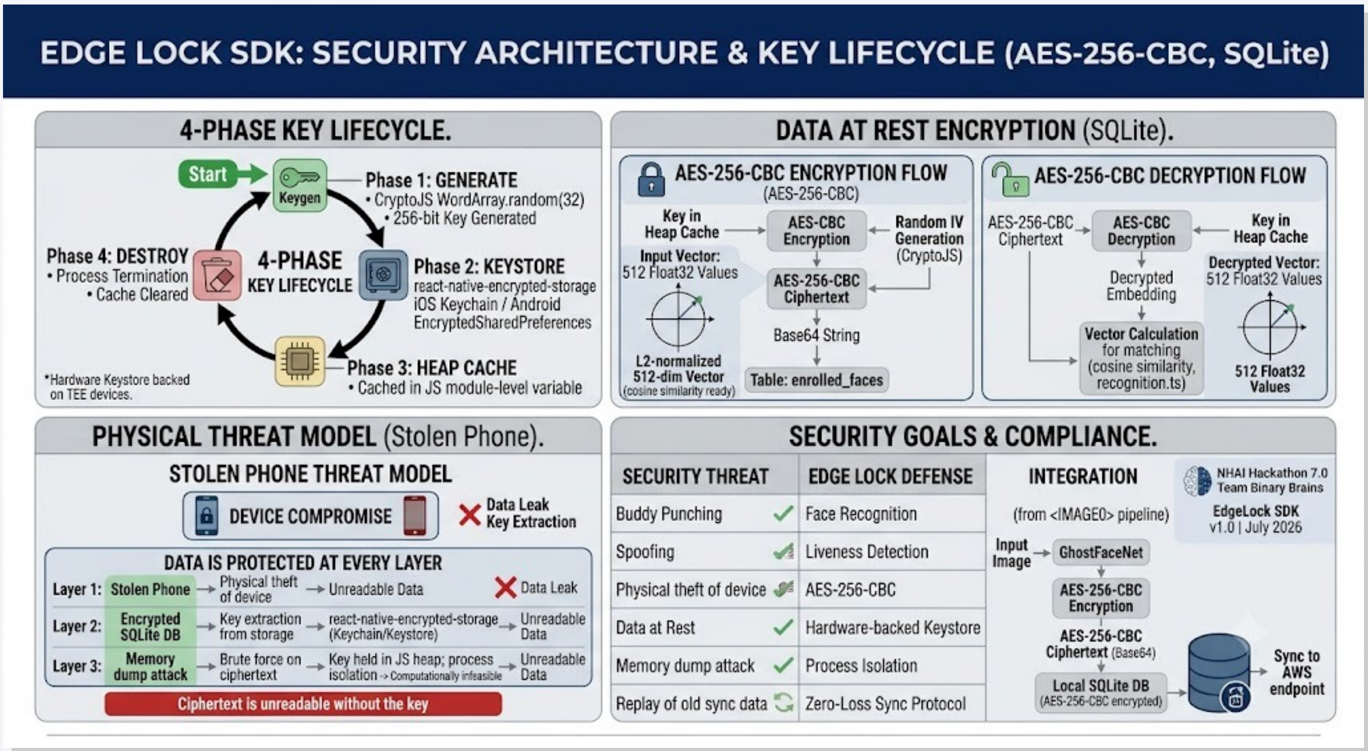
Eye Aspect Ratio (EAR) — Blink Detection State Machine



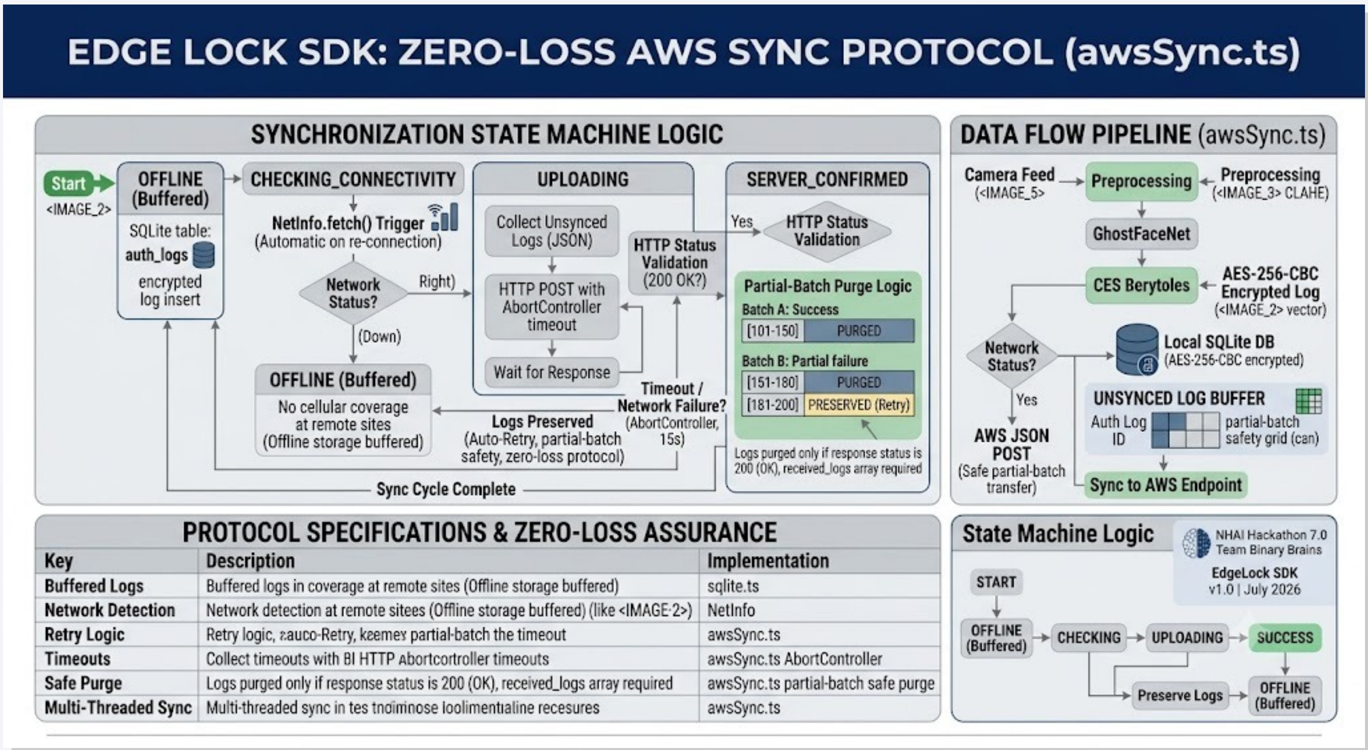
Head Yaw Calculation — Head Turn Detection State Machine



Z-Coordinate Standard Deviation — Passive 3D Spoof Detection



AES-256-CBC Security Architecture & 4-Phase Key Lifecycle



Zero-Loss AWS Sync Protocol — Offline-to-Cloud Pipeline

THANK YOU

Team Binary Brains | NHAI Hackathon 7.0

React Native + Expo SDK 50 | GhostFaceNet INT8 | AES-256-CBC | Zero-Loss Sync

Questions?